

11.24 Reconsider Example 11.2 on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(\text{PRICE})$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?
- Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.
- Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.
- In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

a.

##	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-0.35956847	0.07915415	-4.5426355	1.501702e-05
## mon	-0.10824827	0.10338898	-1.0470000	2.975268e-01
## tue	-0.06609207	0.10103460	-0.6541528	5.144573e-01
## wed	-0.04927356	0.10377370	-0.4748174	6.359127e-01
## thu	0.03934669	0.10070504	0.3907122	6.968092e-01
## stormy	0.44633866	0.07921108	5.6348008	1.508139e-07
## mixed	0.23687528	0.07851343	3.0170036	3.209729e-03

since $p\text{-value of mixed} = 3.209 \times 10^{-3} < 0.05 \Rightarrow$ we reject $H_0 (\pi_{\text{mixed}} = 0)$
and Hence it is statistically significant at the 5% Level

```
##
## Linear hypothesis test:
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##   Res.Df    RSS Df Sum of Sq   F    Pr(>F)
## 1     106 15.876
## 2     104 12.115   2    3.7604 16.14 7.857e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$F\text{-value} = 16.14 > 10$

$\rightarrow$ reject $H_0: \pi_{\text{stormy}} = \pi_{\text{mixed}} = 0$

(b)

```
##           Estimate Std. Error    t value    Pr(>|t|)
## (Intercept)  8.54023437  0.1566081  54.53250335  7.415812e-79
## lprice      -0.93014092  0.3533995  -2.63198161  9.768165e-03
## mon         -0.01190203  0.2083710  -0.05711941  9.545586e-01
## tue         -0.52583152  0.2022954  -2.59932461  1.068445e-02
## wed         -0.56261965  0.2069581  -2.71851907  7.673542e-03
## thu          0.09987059  0.2028454   0.49234827  6.235011e-01
## attr(,"df")
## [1] 105
## attr(,"nobs")
## [1] 111
```

$9.7 \times 10^{-3} ***$
0.95
 $0.0106 ***$
 $7.6 \times 10^{-3} ***$
0.62

TABLE 11.5

2SLS Estimates for Fish Demand

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	8.5059	0.1662	51.1890	0.0000
ln(PRICE)	-1.1194	0.4286	-2.6115	0.0103
MON	-0.0254	0.2148	-0.1183	0.9061
TUE	-0.5308	0.2080	-2.5518	0.0122
WED	-0.5664	0.2128	-2.6620	0.0090
THU	0.1093	0.2088	0.5233	0.6018

$\ln(\text{price}) : -1.1194 \rightarrow -0.9301$ 價格單性↑

TUE, WED → is significant

MON, THU → isn't significant

(c)

```
## Sargan      df p.value
## 0.7722  1.0000  0.3795
```

p-value = 0.3795 > 0.05

⇒ do not reject H_0 (IV is effective)

(d)

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      108 12.408
## 2      104 12.115  4    0.29256 0.6279 0.6437
```

p-value = 0.6437 > 0.05

so we do not reject

$H_0: \text{MON} = \text{TUE} = \text{WED} = \text{THU} = 0$

→ we have no

confidence to estimate
the supply equation with
2SLS.

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?
- Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?
- Estimate the price elasticity of demand "at the means" using the results from (b).
- Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.
- What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?
- Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

a. D: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$

$$\Rightarrow \alpha_2 P_i = Q_i - \alpha_1 - \alpha_3 PS_i - \alpha_4 DI_i - e_{di}$$

$$\Rightarrow P_i = \frac{1}{\alpha_2} Q_i - \frac{\alpha_1}{\alpha_2} - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di}$$

' $\alpha_2 < 0, \alpha_3 > 0, \alpha_4 > 0$, $\rightarrow$ the coef of $\begin{cases} Q_i < 0 \\ PS_i > 0 \\ DI_i > 0 \end{cases}$

S: $Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$

$$\Rightarrow P_i = \frac{1}{\beta_2} Q_i - \frac{\beta_1}{\beta_2} - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si} \quad \text{' } \beta_2 > 0, \beta_3 < 0 \quad \Rightarrow \text{coef of } \begin{cases} Q_i > 0 \\ PF_i > 0 \end{cases}$$

b.

demand

```
##               Estimate Std. Error   t value    Pr(>|t|)
## (Intercept) -11.428407  13.591606 -0.840843  4.081026e-01
## q           -2.670519   1.174955 -2.272869  3.153505e-02
## ps           3.461081   1.115572  3.102517  4.582228e-03
## di          13.389921   2.746707  4.874899  4.675236e-05
## attr(,"df")
## [1] 26
## attr(,"nobs")
## [1] 30
```

$Q < 0, ps > 0, DI > 0$

and all p-value $< 0,05$

$\rightarrow$ significantly different from 0 (reject H_0)

supply

```
##           Estimate Std. Error   t value    Pr(>|t|)
## (Intercept) -58.798223  5.8591611 -10.03526 1.316492e-10
## q           2.936711  0.2157717  13.61027 1.320944e-13
## pf          2.958486  0.1559638  18.96905 3.879687e-17
## attr(,"df")
## [1] 27
## attr(,"nobs")
## [1] 30
```

$q > 0$, $pf > 0$

and all p-value < 0.05

→ significantly different from 0
(reject H_0)

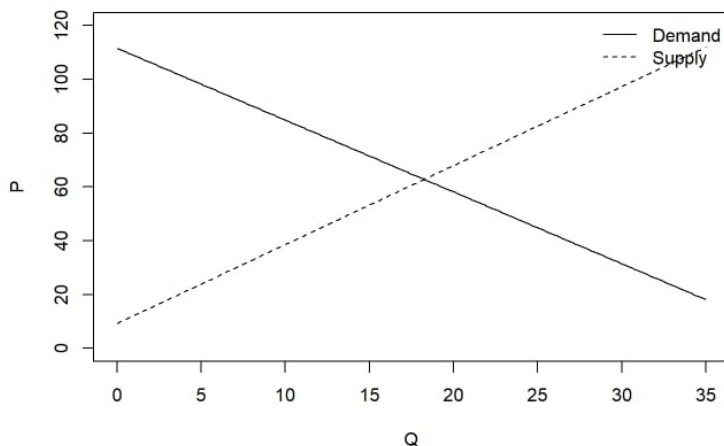
C,

```
##           Pbar           Qbar elasticity.q
##           62.7240          18.4583         -1.2725
```

D,

```
## D_intercept D_slope_Q S_intercept S_slope_Q
##           111.5801        -2.6705           9.2470           2.9367
```

Truffle Demand and Supply



e,
equilibrium values of P, Q in (d):

##	P_eq	Q_eq
##	62.8426	18.2502

predict equilibrium value:

##	Q_rf	P_rf
##	18.2604	62.8154

##		Q	P
##	structural_2SLS	18.2502	62.8426
##	reduced_form	18.2604	62.8154
##	difference	-0.0102	0.0272

→ 兩者結果接近

f, (use OLS)

supply

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	-52.8763	5.0238	-10.5252	0
##	q	2.6613	0.1712	15.5436	0
##	pf	2.9217	0.1482	19.7150	0

p-value < 0.05

→ coef are significantly different from 0

Demand

##		Estimate	Std. Error	t value	Pr(> t)
##	(Intercept)	-13.6195	9.0872	-1.4987	0.1460
##	q	0.1512	0.4988	0.3032	0.7642
##	ps	1.3607	0.5940	2.2909	0.0303
##	di	12.3582	1.8254	6.7701	0.0000

→ not significantly

]→ significantly

compare to (b)

demand (b)

## q	-2.670519
## ps	3.461081
## di	13.389921

q	0.1512
ps	1.3607
di	12.3582

OLS x
OLS 估計
OLS 較小

supply (b)

q	2.936711
pf	2.958486

q	2.6613
pf	2.9217

OLS 較小
接近